

16-bit Security Controller – V02

Qualification Report

About this document

Scope and purpose

This document provides information about the results of the product(s) qualification.

Intended audience

This Qualification Report is intended for customers of Infineon Technologies and certification parties. The data contained in this document is exclusively intended for technically trained staff.

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General Information

1 General Information

Table 1 General Information

Target Application	SOLID FLASH™ Controller for Security Applications
Wafer Process Technology	C65FL
Wafer Diameter	300 mm
Wafer Production Site(s)	Taiwan Semiconductor Manufacturing Company, Ltd., Fab 14, Tainan, Taiwan R.O.C.
PCN	-

1.1 Applied Standards

- JESD47
- ISO 7816
- ISO 10373

2 Family Qualification with Generic Data from Reference Parts / Design / Products

Table 2 Family Qualification

	Product (Identifier)	Wafer Technology	Design Step
Part to be Qualified	16-bit Security Controller-V02	C65FL	G13
1. Reference Part A	16-bit Security Controller-V02	C65FL	G12
2. Reference Part B	16-bit Security Controller-V02	C65FL	G11
3. Reference Part C	16-bit Security Controller-V01	C65FL	H11
4. Reference Part D	SLx 14MCOxxxK2	C65FL	G11

Reference part types / products utilize the same fabrication process, design rules, and similar circuits.

Qualification Result Overview

3 Qualification Result Overview

Table 3 Qualification Result Overview

Qualification Test	Test Conditions	# Lots	Sample Size / Lot	Part to be Qualified	Reference Part			
					Part A	Part B	Part C	Part D
Electrical Characterization	Acc. product specification $T_a = -40\text{ °C} \dots 105\text{ °C}$	3 ¹⁾	≥ 10	Pass	---	---	---	---
System Characterization	Acc. product specification $T_a = -40\text{ °C} \dots 105\text{ °C}$	3 ¹⁾	≥ 10	Ref. part A	Pass	---	---	---
High Temperature Operating Life	$T_a = 125\text{ °C}$, 1000h $V_{CC\text{int.}} = V_{CC\text{int. max}}$ JESD22-A108	3	77	Ref. part B, C	---	Pass (1 lot)	Pass (2 lots)	---
Electrostatic Discharge: Human Body Model	4 kV JESD22-A114	1	20	Pass	---	---	---	---
Electrostatic Discharge: Charged Device Model	750 V JESD22-C101	1	3	Pass	---	---	---	---
Latch Up	Class II JESD78E	1	10	Pass ²⁾	---	---	---	---
NVM Cycling Endurance per Cell	Write/Erase cycles: 200,000 ³⁾ , $T = 25\text{ °C}$ IFX confidence level test flow	3	≥ 100	Ref. part C	---	---	Pass	---
NVM Data Retention	250 °C, 6 h IFX confidence level test flow	2	≥ 100	Ref. part C	---	---	Pass	---
UV Light Exposure	254 nm, 0.5 h, 15 Ws/cm ² ISO 10373-1	1	5	Ref. part D	---	---	---	Pass
X-Ray Exposure	100 kV / 0.1 Gy ISO 10373-1	1	5	Ref. part D	---	---	---	Pass
Static Magnetic Field Exposure	220 kA/m ISO 10373-1	1	5	Ref. part D	---	---	---	Pass

¹⁾ Process Window Lot(s)

²⁾ Immunity Level A

³⁾ 200,000 erase operations per page, 1.6 million block write operations per page (8 blocks per page)

4 Qualification of Delivery Forms

For qualification of the delivery forms please refer to the corresponding qualification reports.

These documents are available at our Customer Service Center.

Please contact: Infineon Technologies AG, Neubiberg, Germany, Chip Card & Security, e-mail:
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Revision History

Revision History

Major changes since the last revision

Page or Reference	Description of change
Revision 1.0, 2017-01-17	
All	First edition

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Document reference

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